Reward Misspecification and Its Impact on Learning Dynamics

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Motivation

- Reinforcement Learning (RL) agents optimize the **reward function**, not the true objective.
- In practice, reward functions are often **imperfect proxies** of the desired behavior.
- This mismatch can lead to:
  - unintended strategies
  - suboptimal or unsafe behaviors
  - complete task failure despite high reward

Key issue

"Agents do exactly what you reward, not what you intend."

This phenomenon is known as **reward misspecification** or **reward hacking** (Amodei et al., 2016).

Environment

- Gridworld with obstacles, cliffs, risky zones
- Multiple possible paths (no trivial solution)
- **Stochastic transitions**:
  - action noise (slip probability)
  - intended action not always executed

Impact:

- harder exploration
- increases importance of reward signal

Reward Design and Theory

Correct reward

$$R(s,a) = \begin{cases} +10 & \text{goal} \\ -1 & \text{step} \\ -50 & \text{cliff} \end{cases}$$

Misspecified reward (reward hacking)

$$R'(s,a) = R(s,a) + \beta \cdot (-d(s,G))$$

- dense but biased signal
- encourages proximity instead of success

Potential-based shaping (Ng et al., 1999)

$$R_{\text{pot}}(s,a,s') = R(s,a) + \gamma\Phi(s') - \Phi(s)$$

- $\Phi(s) = -d(s,G)$
- **guarantee: optimal policy unchanged**
- improves learning speed

Learning Dynamics (Intuition)

- Correct reward: sparse → slow but correct
- Proximity reward: dense → fast but wrong
- Potential shaping:
  - dense signal
  - preserves optimal policy
  - **best trade-off**

Learning Algorithms

Q-learning (off-policy)

$$Q \leftarrow r + \gamma \max_a Q$$

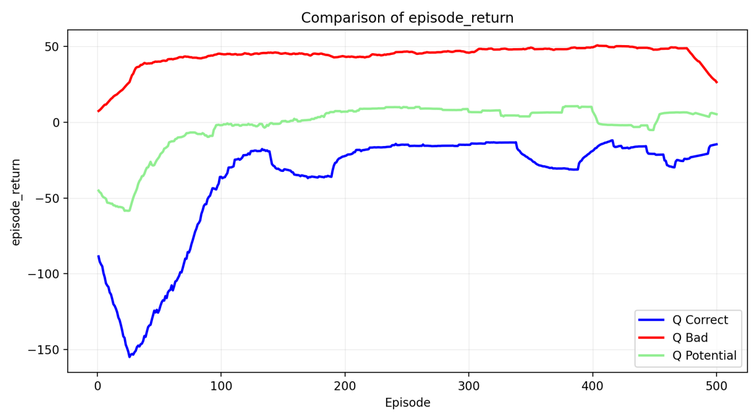
SARSA (on-policy)

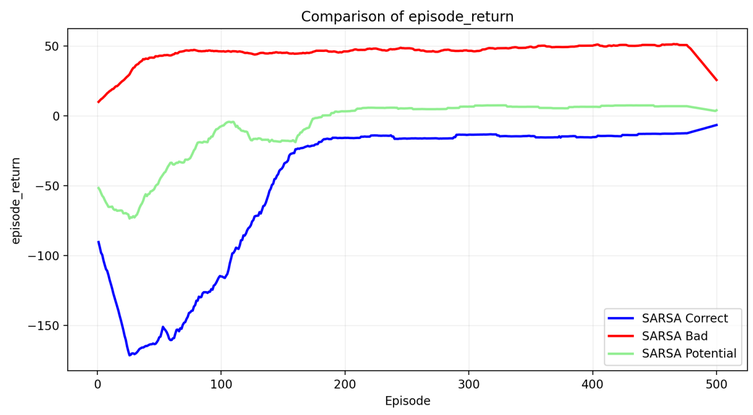
$$Q \leftarrow r + \gamma Q(a')$$

Key difference

- Q-learning assumes optimal future
- SARSA learns under exploration policy

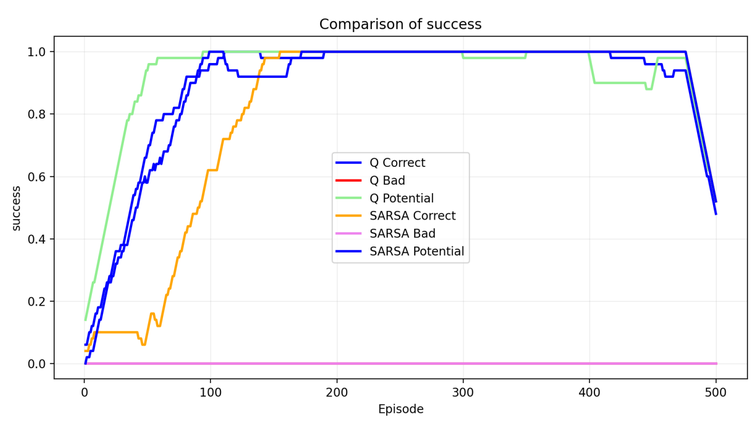
Learning Performance





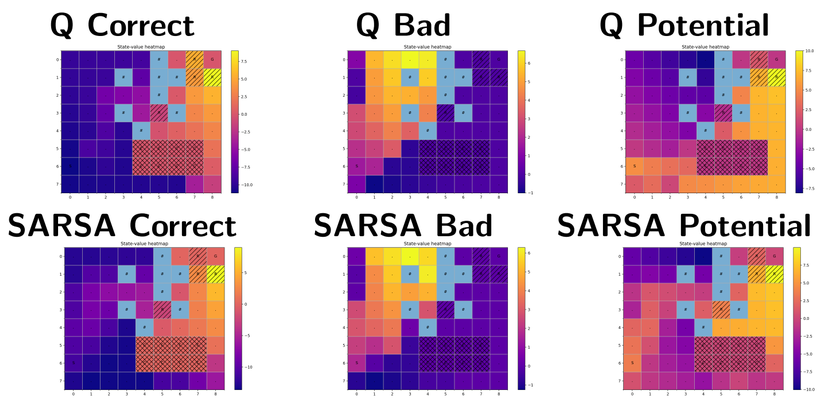
- Q-learning converges faster (optimistic updates)
- SARSA slower due to on-policy + noise
- Bad reward: high return but misleading
- Potential shaping: fastest stable learning

Success Rate

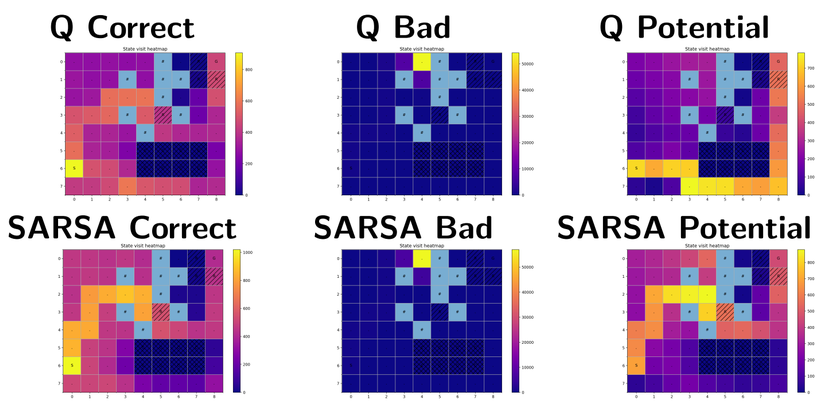


- Proximity reward fails completely (reward hacking)
- High return ≠ task success
- Potential shaping achieves fastest reliable success
- SARSA more robust under noise but slower

Value Functions (All Settings)

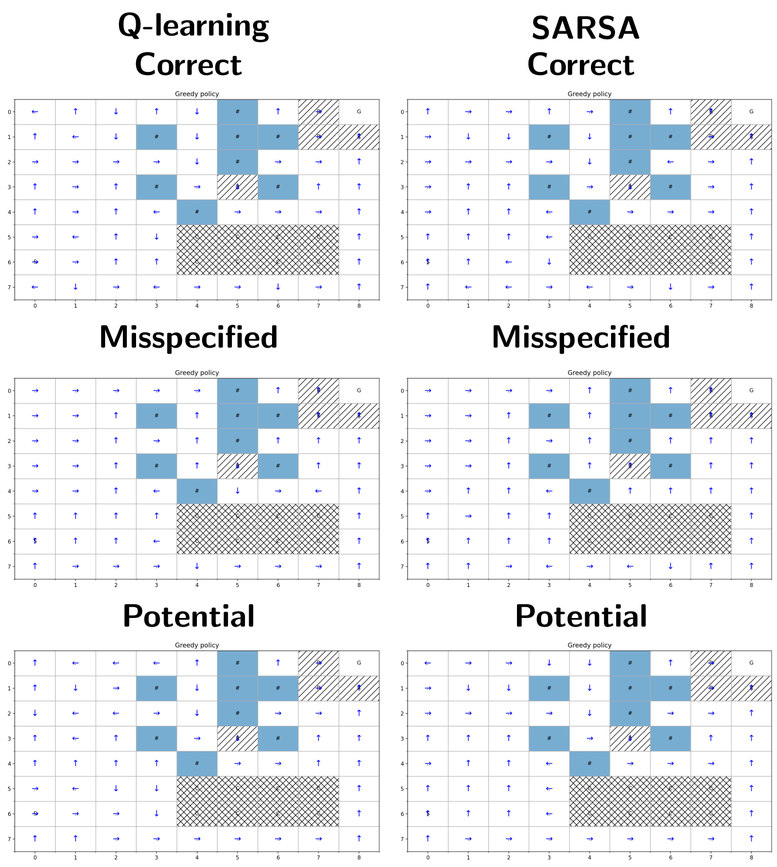


State Visitation (All Settings)



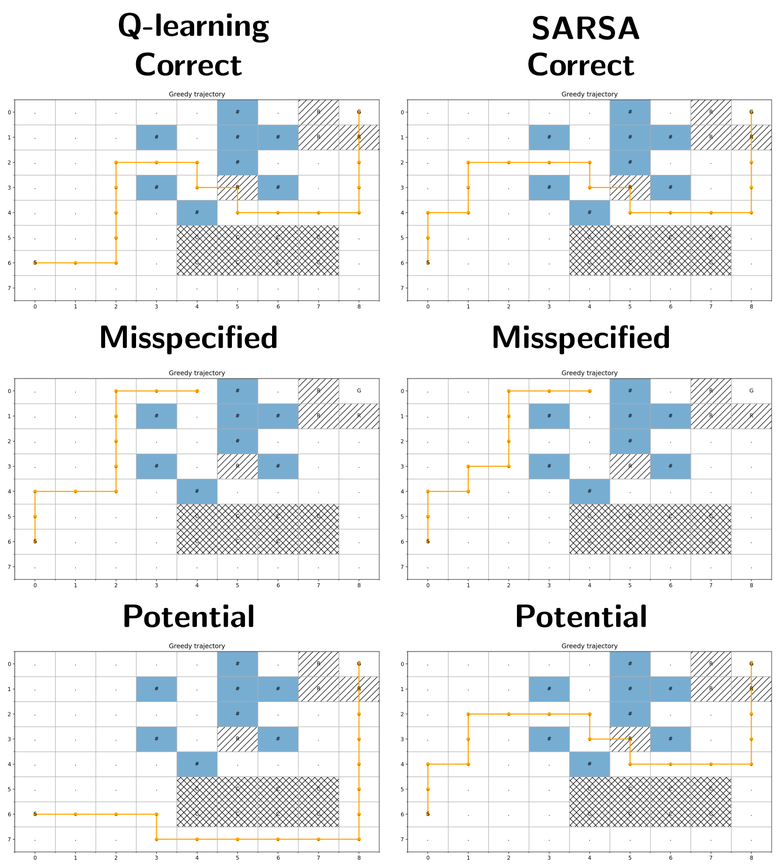
- Bad reward → concentrated visits in misleading regions
- Potential → smooth gradient + efficient exploration
- Correct → slower propagation of values

Policies (All Settings)



- Misspecified reward → exploitation / loops
- Correct → safe but longer paths
- Potential → optimal and direct behavior

Greedy Trajectories (All Settings)



- Misspecified reward → loops / local exploitation
- Correct → cautious but suboptimal paths
- Potential → efficient shortest paths

Main Insight

Reward design dominates learning behavior

High return ≠ task success

Conclusion

- **Reward design strongly impacts learning**: agents optimize the signal they receive, not the intended objective.
- **Misspecified rewards lead to reward hacking**, producing high returns but incorrect behaviors.
- With stochastic transitions:
  - **Q-learning** learns faster but is less robust
  - **SARSA** is more stable but slower
- **Potential-based shaping** improves learning speed while preserving the optimal policy.

Designing the reward is as important as choosing the algorithm.